

European ancestry

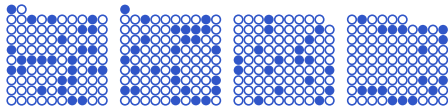
EUR358 in total

FIN 92

TSI 91

CEU 89

GBR 86



— 17.3M SNVs common in either sample

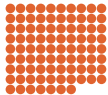
- in random draw 1 of 5 (EUR87)
- not in that draw

Elastic net per gene
weights w_{EUR} on cis-SNPs



DNAJB7, 76 SNPs

~20,800 genes per set



all 87 Yoruba with genotypes,
the same n as each draw



weights w_{YRI} on cis-SNPs

$\alpha = 0.5$, nested cross-validation

Variance decomposition

$$V = w^T D^{1/2} R D^{1/2} w$$

$$\Delta = \log V_{\text{YRI}} - \log V_{\text{EUR}}$$

$$\Delta = \phi_w + \phi_D + \phi_R$$



w weights, D frequency, R LD

Summary-statistic TWAS

European LD reference

ASD

SCZ

BIP

MDD

PTSD

+ GWAS coverage of each model
autism + 4 psychiatric disorders
with large public GWAS

Yoruba, Ibadan
YRI87